

Artificial Intelligence in English as a foreign language learning: A systematic review of technologies, tools, and their impacts

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ABSTRACT

This study undertook a thorough systematic review of the existing literature on artificial intelligence (AI) technologies and AI-based tools for English as a foreign language (EFL) learning and explored the current research state with the goal of identifying areas for enhancement to optimize language learning for EFL learners. A qualitative research method (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) was adopted and three steps were conducted to guide the review. This study examined 708 articles published from 2019 to 2024 and retained 103 in total. The findings show steady increase in literature between 2019 and 2024, predominantly centred on higher education participants, with comparatively less research on elementary, secondary students, and even pre-schoolers. Furthermore, the deep learning method was the most commonly used AI technology, complemented by the most frequent use of AI-based tools, particularly AI-powered chatbots as learning tools. These technologies have been effectively harnessed to bolster EFL learners' integrated, writing, and speaking skills, with overwhelmingly positive impacts on their learning experience. The implications of these findings are twofold: first, there is a pressing need for future research to investigate the application of AI in EFL learning across all age groups; second, further exploration of AI technologies and their integration into diverse pedagogical approaches is warranted. This review highlights gaps in the research and provides recommendations for further studies of AI technologies and AI-based tools in EFL learning for researchers, students, and teachers.

1. Introduction

Artificial intelligence (AI) embodies the machine's ability to delve into and advance theories, procedures, technologies, and application systems employed to imitate, extend, and enhance human intelligence (Imberman, 2004). It involves reforming numerous sectors within contemporary society, including the guidance of English as a foreign language (EFL; Yuan, 2023). The integration of AI technologies, such as machine learning and deep learning, has garnered international interest and has become increasingly significant in the realm of education worldwide (Huang, 2021; Wang et al., 2021). Over the last decade, the focus on EFL has intensified, with studies indicating that students increasingly rely on the progression of computer science and technology, thereby enhancing their computational thinking and abilities (Chen et al., 2020; Zhou, 2023). These technologies are capable of offering learners customized study materials, increasing their interest and motivation (Ying, 2023), and meeting individual learner needs through personalized feedback and assessment (Bhutoria, 2022; Chen et al.,

2021). Specifically, AI-based tools, particularly advanced chatbots, are becoming prevalent in language learning, offering regular dialogues, evaluation, and feedback (Geng, 2021).

However, the integration of AI technologies also presents new challenges such as superficial communication, feedback, and the possible failure to develop higher-order language learning competencies like debate and problem-solving, which are critical for advanced language learning (Fryer et al., 2020; Lin & Chang, 2020; Xu et al., 2021; Zhai & Wibowo, 2023). As the evolution and improvement of AI technologies and AI-based language learning tools continue, language instructors still seek to determine the definitive value and efficacy of such technologies in language learning contexts, although various issues have been raised and are still pending answers (Kim et al., 2021). For instance, the effectiveness of AI chatbots for students with various levels of proficiency in English learning has been debated, with some studies arguing that beginners may not benefit from such technologies, while others argue that students at all levels may benefit from such technologies (Jiang, 2022). Other studies have raised issues relating to the

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hardware and software constraints faced in the Automatic Speech Recognition (ASR) technologies, which may hinder students' learning experiences (Crescenzi-Lanna, 2020).

Despite these challenges, empirical studies have demonstrated the potential of AI technologies to enhance various aspects of EFL learning. In vocabulary acquisition, optical character recognition and image recognition technologies, such as Google Translate and Android mobile learning applications, have been shown to support vocabulary development (Hsu et al., 2023). In writing skills, automated evaluation and feedback systems, including Grammarly and ChatGPT, have demonstrated effectiveness in increasing writing efficiency while reducing anxiety levels (Wang et al., 2022; Yan, 2023). In speaking skills, AI-powered tools such as Mondly and voice assistants have been associated with increased willingness to communicate and reduced public speaking anxiety (Chen, Koong, & Liao, 2022; Rahman & Tomy, 2023; Yuan, 2023). These findings underscore the transformative potential of AI technologies across multiple dimensions of EFL learning.

To establish a clear conceptual foundation for this review, it is essential to distinguish between three interrelated but distinct constructs. AI technologies refer to the underlying computational techniques and algorithms, such as machine learning, natural language processing (NLP), ASR, and computer vision, that enable intelligent behaviors. AI-based tools, in contrast, represent the practical applications and software systems that utilize these underlying AI technologies to perform specific functions, such as writing assistants (e.g., Grammarly), pronunciation trainers, or intelligent tutoring systems. Within this category, AI chatbots constitute a distinct subset: they are conversational agents that leverage generative large language models or advanced NLP to facilitate dynamic, context-dependent dialogue. This review deliberately excludes rule-based chatbots, as they lack the adaptive, generative capabilities central to the current wave of AI innovation in education.

More critically, the expanding body of empirical research in this area remains fragmented across technologies, learner groups, and methodologies, hindering the development of a cohesive understanding of AI's role in EFL learning. To address the identified gaps and achieve the aim of providing a comprehensive, structured synthesis of the field, this systematic review is guided by the following research questions. Distinct from prior systematic reviews that have focused on single technologies or specific learner populations, this study adopts a two-tier analytical framework distinguishing foundational AI technologies from applied AI-based tools, synthesizes findings across the full spectrum of language skills, and interprets results through multiple theoretical lenses. Each question is designed to map a specific dimension of the current research landscape and directly respond to the fragmentation and inconsistencies. First, to quantify and characterize the body of literature, RQ1 examines publication trends, participant demographics, and methodological approaches. This mapping is essential to identify which learner groups and research designs have been prioritized or overlooked, thereby addressing the gap in a cohesive understanding of the field's scope: What are the publication trends, the education levels of the participants, and the research methodology in the studies on AI technologies affecting EFL learners from 2019 to 2024? Second, to identify and categorize the predominant AI technologies deployed in EFL research, RQ2 directly addresses the fragmentation across different tools and systems by establishing a clear taxonomy of what constitutes "AI technologies" in this context and determining their relative frequency of use: What are the most common AI technologies used for EFL learning? Finally, moving beyond mere technological description to consolidate evidence on efficacy, RQ3 is designed to synthesize the reported outcomes and effects of these technologies, thereby directly responding to the pressing need for a clearer understanding of the impact of AI on EFL learning outcomes: What are the effects developed in research on AI technologies and AI-based tools for EFL learning?

2. Literature review

2.1. Previous systematic reviews and research gaps

Several systematic reviews have synthesized specific facets of this emerging field, providing foundational insights into the applications and effectiveness of AI in EFL learning. Sharadgah and Sa'di (2022) conducted a systematic review of AI applications in English language teaching and learning from 2015 to 2021, focusing primarily on the overall effectiveness of AI tools across various language skills. Their findings indicated a promising future for AI in EFL learning, with positive effects reported across multiple language skills, translation, assessment, and learner attitudes. Importantly, their review revealed that higher education participants were the most frequently sampled population, and mixed methods were the most commonly employed research design. However, the review covered literature only up to 2021, predating the widespread adoption of generative AI technologies such as ChatGPT. Zhai and Wibowo (2023) specifically examined AI dialogue systems for enhancing EFL students' interactional competence in university settings, identifying six dimensions and 25 sub-dimensions that influence the application of such systems. Their review highlighted the potential of dialogue systems to facilitate interactive practice but noted that components of debate and problem-solving skills were often overlooked in system design. More recently, Kundu and Bej (2025) explored the transformation of EFL teaching through AI, synthesizing empirical studies to identify general trends and pedagogical implications in school-based contexts. Their review covered literature from 2020 to 2024 and focused exclusively on primary and secondary education settings. Aliakbari et al. (2025) specifically investigated AI-powered chatbots for English conversation practice through a controlled experimental design, contributing empirical evidence on the comparative efficacy of chatbot-mediated versus peer-to-peer practice.

Despite these valuable contributions, three critical gaps remain unaddressed in the existing review literature. First, the literature remains fragmented across technologies and learner populations. Sharadgah and Sa'di covered a broad range of AI applications but did not distinguish between different types of AI, while subsequent reviews each focused on a single technology or educational level, such as dialogue systems in universities, school-based EFL, or chatbots. No single review has integrated findings across the full spectrum of AI applications and learner populations. Second, existing reviews have not systematically distinguished between foundational AI technologies (e.g., machine learning, NLP, ASR) and applied AI-based tools (e.g., chatbots, automated writing evaluation systems). Sharadgah and Sa'di listed various AI approaches without establishing a hierarchical taxonomy, and Kundu and Bej similarly presented NLP, chatbots, and intelligent tutoring systems as parallel categories. This conceptual ambiguity hinders the development of a coherent framework for understanding AI's role in EFL learning. Third, previous syntheses have largely been descriptive, cataloguing positive effects across language skills without providing theory-driven explanations for observed patterns. While these reviews effectively documented that AI tools improve various language skills, they did not systematically explain why certain technologies are more effective for particular skills or how AI tools interact with established second language acquisition processes. Sharadgah and Sa'di, for instance, reported positive effects across multiple dimensions but did not interpret these findings through theoretical lenses. Fourth, beyond these conceptual and methodological gaps, empirical evidence on AI's effectiveness remains fragmented. While some studies have demonstrated positive effects of automated feedback on writing quality (Stevenson & Phakiti, 2019) and improvements in critical thinking through AI-based tools (Ibna Seraj & Oteir, 2022), there is a lack of overall evidence on the long-term impact of these technologies on writing achievement. Similarly, challenges such as cognitive overload and the limitations of ASR technology in supporting speaking skills remain under-researched (De Vries et al., 2015; Liu & Li, 2022; Young & Wang, 2014). These

inconsistencies further underscore the need for a comprehensive synthesis (Sun & Rui, 2025; Zhang & Huang, 2024).

2.2. AI technologies and AI-based tools in EFL learning: A two-tier taxonomy

The application of AI in EFL learning rests on a foundational distinction between AI technologies, the underlying computational techniques and algorithms, and AI-based tools, the practical applications that operationalize these techniques for language learning purposes. This two-tier taxonomy directly addresses RQ2 by providing a structured framework for categorizing and comparing the technologies and tools reported in the reviewed literature.

At the technology level, four core AI techniques dominate the EFL research landscape. NLP enables machines to understand, interpret, and generate human language, making it essential for automated writing evaluation, chatbot dialogue systems, and vocabulary recommendation engines (Jiang, 2022; Lew et al., 2024). ASR converts spoken language into text, facilitating pronunciation assessment, fluency analysis, and interactive speaking practice. ASR systems powered by deep learning have demonstrated particular effectiveness in providing real time, phoneme level feedback, although challenges persist with non-native accents and noisy environments (Crescenzi-Lanna, 2020; Jiang et al., 2023). Computer vision technologies allow learners to capture visual data and receive linguistic output, and have been effectively harnessed for vocabulary acquisition and contextualized writing support, particularly for younger learners (Hsu et al., 2023). Deep learning architectures including neural networks (NN), recurrent neural networks (RNN), and convolutional neural networks (CNN) serve as the infrastructural backbone for these technologies, enabling the sophisticated pattern recognition that underpins contemporary AI-based tools (Gao, 2021; Liu, 2023).

At the tool level, AI based applications translate these computational capabilities into practical learning resources. Based on the reviewed literature, six functional categories emerge. AI-powered chatbots (e.g., ChatGPT, custom built conversational agents) represent the most frequently researched tool category. Distinguished from rule based chatbots by their use of generative large language models or advanced NLP, these agents facilitate dynamic, context dependent dialogue and support multiple language skills simultaneously, providing conversational practice for speaking fluency, generating writing feedback, and offering vocabulary explanations on demand (Chen, Hsu, et al., 2022; Kim et al., 2021; Yuan, 2023). This review deliberately excludes rule based chatbots, as they lack the adaptive, generative capabilities central to current AI innovation in education. Automatic evaluation and feedback systems (e.g., Grammarly, automated writing evaluation platforms) provide immediate, targeted feedback on grammar, vocabulary, organization, and content by drawing learners' attention to specific errors in their written output (Dizon & Gayed, 2021; Wang et al., 2022). Intelligent tutoring systems offer structured, curriculum aligned instruction with adaptive feedback, typically designed for specific learning objectives such as reading comprehension or vocabulary acquisition, and often incorporate learner modeling to personalize instruction (Ni & Cheung, 2023; Xu et al., 2022, 2024). Speech and image recognition systems leverage ASR and computer vision technologies to support pronunciation training and vocabulary acquisition, providing real time, multimodal feedback that makes them particularly suitable for younger learners and self-directed practice (Chen, Koong, & Liao, 2022; Hsu et al., 2023). Computer aided learning and teaching technologies encompass broader platforms that support various learning activities beyond targeted feedback, often combining multiple AI technologies to create integrated learning environments (Ouyang et al., 2024; Zhou, 2023). Translation applications (e.g., Google Translate) support comprehension and production by providing instant cross linguistic equivalents, though concerns about overreliance and accuracy persist (Kim & Cha, 2023; Ouyang et al., 2024).

The dominance of AI chatbots reflects the field's emphasis on interactive, output oriented learning environments. This pattern, together with the distribution of other technologies and tools, provides the empirical grounding for the theoretical interpretation that follows in Section 2.3, where adaptive learning theory, the Output Hypothesis, and Sociocultural Theory are applied to explain these observed patterns.

2.3. Theoretical frameworks for AI-mediated language learning

This section presents five theoretical perspectives that collectively constitute the analytical framework for this review. These theories function as conceptual tools for organizing, interpreting, and critically examining the patterns observed across the reviewed literature. These theories address three interrelated questions: how AI personalizes instruction, how it facilitates language processing, and how it affects learner motivation and social interaction. The five frameworks map onto the three research questions as follows: adaptive learning theory primarily addresses RQ2 (technologies and tools); the Output Hypothesis and Noticing Hypothesis guide RQ3 (learning outcomes); Sociocultural Theory frames RQ1 (participant demographics); and Self-Determination Theory operationalizes the interpretation of affective outcomes in RQ3 while also identifying research gaps.

Adaptive learning theory emphasizes personalized learning experiences tailored to individual learner needs (Herlo, 2012). This framework aligns with the capabilities of AI technologies, which can deliver immediate feedback and adapt learning resources to each learner's unique requirements (Rane et al., 2024). In the context of this review, adaptive learning theory provides a lens for interpreting the patterns of AI technology and tool deployment examined in RQ2. The theory suggests that the field's emphasis on adaptive, individualized instruction has driven the development and adoption of technologies capable of real-time personalization.

Building on adaptive learning theory, two complementary second language acquisition theories explain how AI tools facilitate language acquisition. The Output Hypothesis (Swain & Lapkin, 1995) posits that language production, particularly "pushed output," is essential for developing fluency and accuracy, as it compels learners to process language syntactically rather than merely semantically. AI chatbots are particularly well-suited to facilitate this process, as they require learners to produce output, receive feedback, and modify their responses in real time. Complementing this, the Noticing Hypothesis (Schmidt, 1990) suggests that learners must consciously notice gaps in their interlanguage for development to occur. AI-based feedback systems can trigger this noticing process by providing immediate, targeted feedback that highlights specific errors, a capability that distinguishes AI from delayed or inconsistent human feedback. Together, these two theories guide the interpretation of RQ3 findings. Specifically, they suggest that AI tools supporting output production and error noticing should be most effective for developing speaking and writing skills.

Beyond cognitive processing, two theories address the social and affective dimensions of AI-mediated learning. Sociocultural Theory (Vygotsky, as cited in Hausfather, 1996) emphasizes that learning is mediated through social interaction and occurs within the Zone of Proximal Development. In AI-mediated environments, AI tools can be conceptualized as potential "simulated interlocutors" or "artificial scaffolds" that provide support within the learner's Zone of Proximal Development. However, unlike human interlocutors, AI tools lack genuine intentionality and shared cultural context, suggesting they are best conceptualized as complementary scaffolds rather than replacements for human interaction. This framework informs the interpretation of RQ1 findings, particularly the demographic patterns observed in the reviewed literature. The theory raises critical questions about whether AI tools designed for adult, self-directed learners are developmentally appropriate for younger populations whose language acquisition relies more heavily on concrete, play-based, and socially mediated interactions.

Self-Determination Theory (Ryan & Deci, 2000) provides a motivational framework for understanding the psychological conditions that support optimal learning. The theory identifies three basic psychological needs, autonomy, competence, and relatedness, which must be satisfied for intrinsic motivation to flourish. AI tools have the potential to support all three needs, but the theory also raises concerns about whether excessive dependence on AI-generated feedback may undermine learner autonomy over time. This framework informs the interpretation of affective outcomes reported in RQ3. Specifically, it suggests that AI tools may support learners' intrinsic motivation by offering control over learning pace (autonomy), providing mastery-oriented feedback (competence), and simulating interactive dialogue (relatedness). However, the theory also raises a critical question that the predominantly cross-sectional literature has not yet addressed: whether excessive dependence on AI-generated feedback might, over time, undermine rather than enhance learner autonomy.

In summary, the five theoretical frameworks serve distinct analytical functions. Adaptive learning theory explains the patterns of technology and tool deployment examined in RQ2. The Output Hypothesis and Noticing Hypothesis account for the distribution of learning outcomes across different language skills investigated in RQ3. Sociocultural Theory illuminates the demographic patterns observed in RQ1 and raises developmental appropriateness concerns. Self-Determination Theory guides the interpretation of affective outcomes, particularly regarding how AI tools may satisfy or undermine learners' basic psychological needs for autonomy, competence, and relatedness. By anchoring the subsequent analysis in these theoretical perspectives, this review moves beyond the descriptive cataloguing characteristic of prior systematic reviews (see Section 2.1) and offers a theory-driven interpretation of the field's current landscape.

3. Method

In this research, the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) process was used to ensure the literature's quality and relevance. The PRISMA contains a checklist and

diagram, as illustrated in Fig. 1. This systematic review delineates the exploratory research questions by articles selecting, identifying, and analysing present information (Colquhoun et al., 2014). This study aims to understand the effects of AI technologies on EFL learning via the selected relevant articles. This study uses three steps to conduct a systematic review and analysis: (i) article selection; (ii) article screening and inclusion; and (iii) data coding, extraction, and analysis. Using the three-step method, the related articles are collected and then, the current research focus of this review is outlined. Finally, the study results are presented after evaluating the samples by content analysis.

3.1. Article selection

The first phase of this review is to select articles to identify and retrieve the literature based on the inclusion and exclusion criteria considered as sources. The criteria for collecting target articles are as follows: (a) published in English in a full-text article, (b) related to the application of "AI" or "Artificial intelligence" or "AI technology" or "Intelligent support" and associated with "Effect" or "Effectiveness", (c) associated with "Language Learning" or "English" or "Second Language" or "Foreign Language" or "ESL" or "EFL", (d) in a peer-reviewed journal indexed in Scopus, (e) provided empirical data, and (f) published from January 2019 to December 2024. The selection of search terms was guided by the study's research questions and the need to balance sensitivity (retrieving all relevant articles) and specificity (excluding irrelevant ones). The core concepts identified for the search strategy corresponded directly to the keywords of this review: Artificial intelligence (and its variants, e.g., AI) to capture the technological domain; AI-based tools to focus on applied systems and software implementations; EFL learning (and related terms, e.g., English as a foreign language) to specify the educational context and learner population; and Systematic review as a methodological filter where appropriate in the search process. This combination ensured a comprehensive yet targeted retrieval of literature pertinent to the effects of AI technologies and tools in EFL contexts.

Articles are excluded when they (a) were composed in languages

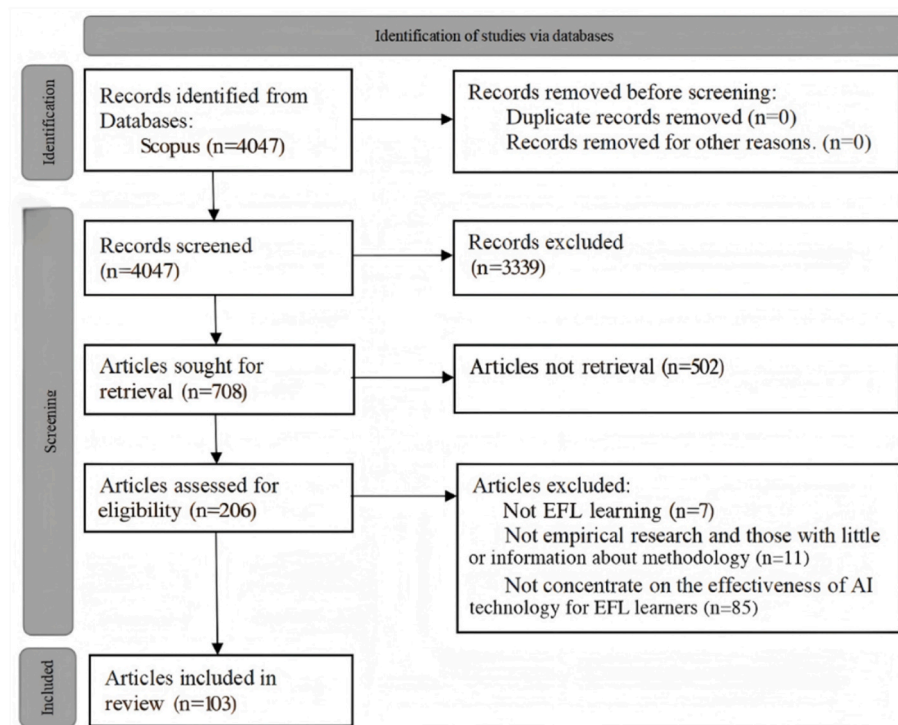


Fig. 1. The PRISMA flowchart.

aside from English, (b) did not concentrate on the application of “AI” or “Artificial intelligence” or “AI technology” or (c) did not focus on English as EFL learning and focused on other foreign language learning, (d) were written in peer-reviewed conference proceedings, book reviews, editorial materials, chapters, and reviews, (e) were not empirical research and those with little or information about methodology, and (f) were published before January 2019 and after December 2024. During this process, the empirical research with a well-designed methodology was considered within the review by referring to the claims of [Firmstone \(2019\)](#). Similarly, editorials and opinion pieces were excluded from this review because of lacking methodological rigor. The inclusion and exclusion criteria are listed in [Table 1](#).

The Scopus was selected because it is recognized as a leading global database, providing a comprehensive index of abstracts from over 5000 publishers across key academic fields, solidifying its role as a foundation for scholarly research. Its selection was further justified for this systematic review for two primary reasons. First, its extensive multidisciplinary scope adequately encompasses the intersecting fields of artificial intelligence, educational technology, and applied linguistics central to our research questions. Second, utilizing a single, comprehensive database enhanced the consistency and replicability of the search process ([Pizzi et al., 2020](#)). While we acknowledge that the exclusive use of Scopus is a potential limitation, as it may not capture all relevant studies indexed in other databases such as Web of Science (WoS), its breadth and relevance to the review’s thematic focus were deemed sufficient to achieve the study’s objectives. This determination was further guided by the acknowledgment of substantial overlap between Scopus and WoS regarding citation coverage, with a reported overlap rate of 88% in the Social Sciences domain, and generally high overlap across most subject areas, particularly in STEM and social sciences ([Martín-Martín et al., 2021](#)), thus rendering Scopus a representative and efficient resource for the current review.

The range of years from 2019 to 2024 was then designated, and a total of 4047 results including 708 potential articles found relevant after applying the inclusion and exclusion criteria as shown in [Fig. 1](#). Apparently, AI technologies have emerged as a highly developed and prolific field of research, and have been increasingly recognized as an innovative instructional instrument worldwide with an abundance of scholarly works published in recent years, particularly after the pandemic. Additionally, in 2019, generative AI reached a significant milestone. The advancements in Deep Learning and Neural Network architectures, particularly the significant strides in the evolution of GPT models, expanded the capabilities of AI in language understanding and generation. This progress enabled more sophisticated applications, such

as improved language translation, content creation, and even automated coding solutions, further solidifying generative AI’s role as a key driver of innovation in the field of artificial intelligence. Therefore, we narrowed our research by considering studies from 2019. Moreover, acknowledging the swift and transformative nature of technology, this study is designed to shed light on AI technologies and AI-based tools for EFL learning. To ensure relevance and to mitigate the risk of encountering outdated digital tools or technological aspects, we have deliberately focused this review on the most recent literature, reflecting the developments and innovations. Nevertheless, this review has meticulously incorporated abundant powerful studies that not only substantiated our findings but also furnished insights to address the research questions.

3.2. Article screening and inclusion

The second phase of this review is the article screening and inclusion procedure to find articles for the main analysis. As reported in [Section 3.1](#), the initial database search yielded 4047 records. After applying the inclusion and exclusion criteria ([Table 1](#)) to the titles, abstracts, and keywords, 3339 records were excluded. The main reasons for exclusion at this stage were that the articles were not centred on the application of AI in EFL learning, belonged to ineligible publication types (e.g., conference proceedings, book chapters, reviews), or did not focus on English as the target foreign language. Consequently, 708 articles remained for full-text scrutiny. During this detailed review, a further 502 articles were excluded after titles, abstracts, and keywords were read to identify empirical studies related to AI technologies and AI-based tools for EFL learners published in journals, thus 502 articles were excluded. Finally, 103 articles were filtered out leaving 103 relevant ones shown in [Appendix](#), after the reading of the remaining articles’ complete texts due to the irrelevance of EFL learning, empirical research, and the effectiveness of AI technologies for EFL learners.

3.3. Coding and analysis

In the third phase, coding categories were set up to respond to the three questions by both deductive and inductive approaches ([Scott & Howell, 2008](#)) and quoted into the coding scheme to emphasize the effects of AI technologies on EFL learning. This coding process was conducted in Microsoft Word and Microsoft Excel. The authors meticulously applied the three-step coding as proposed by [Vollstedt and Rezat \(2019\)](#) based on the Ground theory, ensuring a thorough analysis of the transcripts to identify the AI technologies and AI-based tools and their effects on EFL students. This structured process commenced with open coding, which entailed an initial categorization of the data. The axial coding and selective coding were then followed to further refine the analysis shown in [Appendix](#).

To emphasize the three RQs, the basic parameters, such as supporting study, participants’ educational level, research method, AI technology, AI-based tool, and effect description were considered as coding sources from the 103 selected articles for further analysis. The authors coded the articles as coders manually and 16 articles were first coded based on the basic parameters. Then the coding method and parameters were discussed to confirm with each other before the rest of the selected articles were coded independently. To ensure coding reliability, a formal inter-coder agreement check was conducted. Two authors independently coded a subset of 16 articles (approximately 20% of the total sample) and compared their results. The inter-coder agreement rate was high, indicating strong consistency. Any discrepancies were resolved through in-depth discussion until a full consensus was reached. This process helped to refine the coding scheme and ensure a shared understanding of the categories. Following this calibration, the remaining articles were coded independently by the authors adhering to the established scheme. As the coding table was put into an order according to enhancing skills and other abilities generated by AI technologies for

Table 1
Inclusion and exclusion.

Inclusion	Exclusion
(a) Articles must be published in English in a full-text article,	(a) Articles were composed in languages aside from English,
(b) Articles must be related to the application of “AI” or “Artificial intelligence” or “AI technology” or “Intelligent support” and associated with “Effect” or “Effectiveness”,	(b) Articles did not concentrate on the application of “AI” or “Artificial intelligence” or “AI technology” or “Intelligent support” or associated with “Effect” or “Effectiveness”,
(c) Articles must be associated with “Language Learning” or “English” or “Second Language” or “Foreign Language” or “ESL” or “EFL”,	(c) Articles did not focus on English as EFL learning and focused on other foreign language learning,
(d) Articles must be in a peer-reviewed journal indexed in Scopus,	(d) Articles were written in peer-reviewed conference proceedings book reviews, editorial materials, chapters, and reviews,
(e) Articles must include empirical data, and	(e) Articles were not empirical research or those with little information about methodology, and
(f) Articles have to be published from January 2019 to December 2024.	(f) Articles were published before January 2019 and after December 2024.

EFL learners, some articles could fall into more than one categorization. Thus, an open coding format was used and the information of the research was extracted based on the summarized main results of effect descriptions.

The encoding for each categorization, as shown in [Appendix](#), is described as follows:

1. Participants' educational level: These codes focus on the educational levels of participants in the research and are categorized into elementary, secondary, tertiary, cross-level education, and unspecified and others.
2. Research method: These codes emphasize the importance of research strategies for EFL learners and are fallen into Quantitative, Qualitative, and Mixed Methods.
3. AI technologies and AI-based tools: These codes address different technological components and AI-based tools used and affected for EFL learning.
4. Effect description: These codes highlight the specific outcomes that AI technologies and AI-based tools achieved for EFL learners.

4. Results

4.1. What are the publication trends, the education levels of the participants, and the research methodology in the studies on AI technologies affecting EFL learners from 2019 to 2024? (RQ1)

[Fig. 2](#) presents the number of articles included in this review increasing constantly from January 2019 to December 2024.

[Fig. 3](#) shows the educational level of participants in the reviewed articles. Tertiary education accounts for the highest proportion (75%) in this review with 77 articles, followed by elementary education, unspecified, and others both with 8 articles (8%), secondary education with 7 articles (7%), and cross-levels with 2 articles (2%). The research objects are EFL learners with mainly higher education and the research has been less commonly applied to elementary, secondary students, and even pre-schoolers.

[Fig. 4](#) shows that there are 64 articles (62%) in this review whose research method is quantitative, followed by 33 articles with a mixed method (32%), and 6 articles with a qualitative method (6%).

4.2. What are the most common AI technologies used for EFL learning? (RQ2)

[Table 2](#) displays the distribution of adopted AI algorithms and models in this review. Deep learning (51%) is the most popular, followed by machine learning (8%), models conducted by the algorithms (38%), and Spectrum algorithms (3%).

Meanwhile, 19 articles used deep learning methods, such as recognition, NN, RNN, NLP, CNN combined with NLP, and NLP and recognition. Among the recognition articles, two used automatic speech technology and two used automatic image recognition technology methods. Moreover, 14 articles used models conducted by the algorithms, such as an English-intelligent tutoring teaching model based on the radial basis function (RBF) interpolation and judgment trees, and a learner model performed by the RBF interpolation and judgment trees. Among studies that used machine-learning methods, two employed data mining and one used Spectrum algorithms.

[Table 3](#) shows the distribution of adopted AI-based tools in this review. Based on their ranking, the most commonly used AI-based tool is AI chatbot (specifically those powered by generative AI or advanced NLP models) (56%) which was studied by 44 articles. Nine articles used computer-aid learning technology accounting for 12%; seven used automatic evaluation/feedback systems (9%); and five articles used intelligent tutoring systems (6%). Four articles (5%) used computer-aid teaching technology, translation apps, and speech/image recognition systems. The research volume on the use of VR mobile-assisted tools was the lowest.

4.3. What are the effects developed in research on AI technologies for EFL learning? (RQ3)

The distribution of articles in terms of skills and abilities that are generated from AI technologies for EFL learning is exemplified in [Table 4](#). There are overwhelmingly positive results over 103 articles in this review. The proportion of articles regarding enhancing EFL learners' integrated skills (25 articles) and writing skills (20 articles) is nearly 44%, followed by enhanced speaking skills (18 articles, 17%), enhanced cognition (5 articles, 5%), and being willing to engage in EFL learning (11 articles, 11%), input and comprehension skills (6 articles, 6%), enhanced vocabulary acquisition skills (8 articles, 8%), reduced speech-related anxieties (6 articles, 6%), enhanced translation Skills (3 articles, 3%), enhanced listening Skills (1 articles, 1%).

In terms of language skills and abilities obtained from using AI

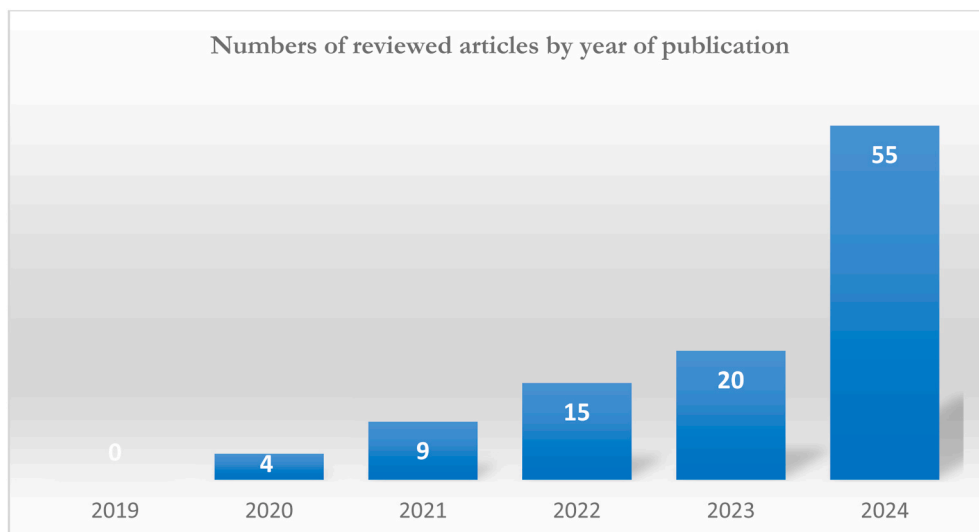


Fig. 2. Numbers of reviewed articles by year of publication.

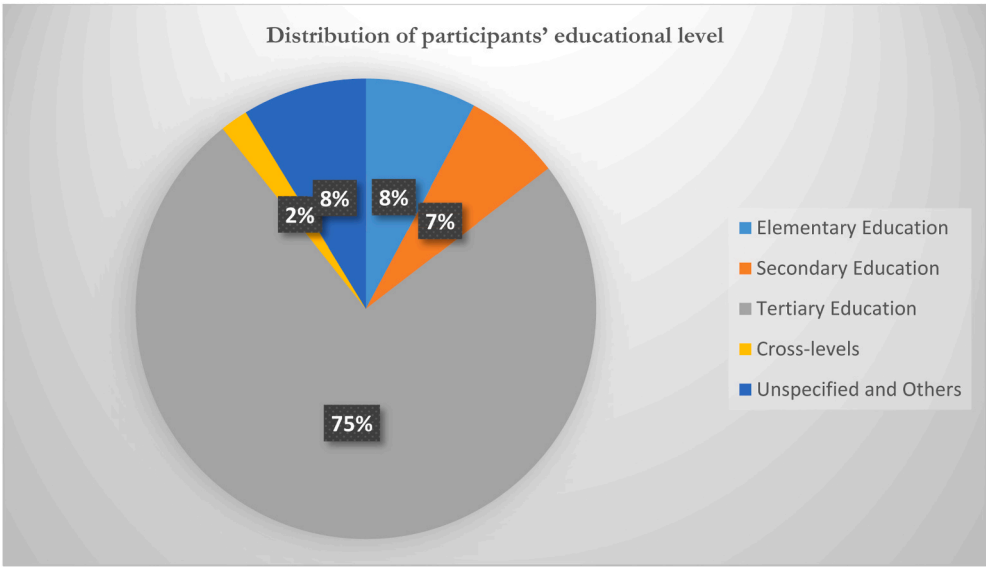


Fig. 3. Distribution of participants' educational level.

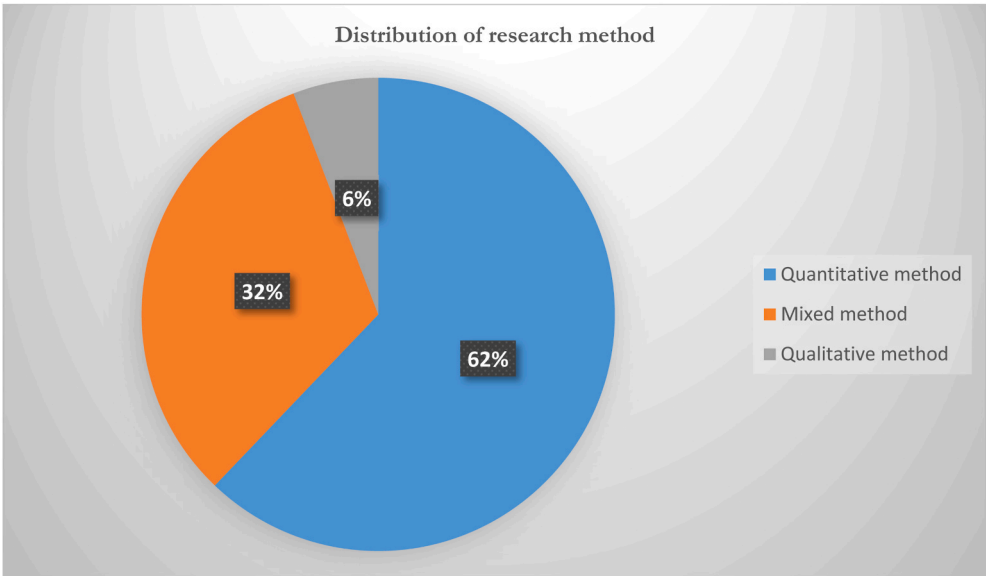


Fig. 4. Distribution of research method.

Table 2
Distribution of adopted AI algorithms and models.

Primary Category	Specific Technology	Number of Articles
Machine learning (Deep Learning) (19)	Recognition	4
	NN	2
	RNN	1
	NLP	7
	RNN, CNN, and NLP	3
	NLP and recognition	2
Machine learning (Non Deep Learning) (3)		3
Models conducted by the algorithms (14)	An English-intelligent tutoring teaching model	11
	A learner model	3
Spectrum algorithm (1)		1

technologies, most studies centered on enhancing integrated English

skills. This was followed by enhancing writing skills, speaking proficiency, cognition, and input and comprehension. Enhancing vocabulary acquisition skills, reducing speech-related anxieties, being willing to engage in EFL learning, and enhancing English translation skills were the least studied abilities in this review.

5. Discussion

5.1. The publication trends, the education levels of the participants, and the research methodology in the studies on AI technologies affecting EFL learning from 2019 to 2024 (RQ1)

The findings of RQ1 reveal a concentrated yet rapidly evolving research landscape, characterized by significant demographic and methodological patterns that warrant a theory-informed interpretation. The steady increase in publications from 2019 to 2024 aligns with the rapid advancements in generative AI (e.g., GPT models) and the widespread adoption of online learning tools during the COVID-19 pandemic.

Table 3

Distribution of adopted AI-based tools.

AI chatbot	Computer-aided learning technology	Automatic evaluation/feedback system	Intelligent tutoring system	Computer-aided teaching technology	Translate app	Speech/image recognition system	Others
44	9	7	5	4	4	4	1

Table 4

Distribution of research based on the skills of EFL learners.

Skills	Number of Articles
Enhanced Integrated English Skills	25
Enhanced Writing skills	20
Enhanced Speaking Skills	18
Enhanced Cognition	5
Being Willing to Engage in EFL Learning	11
Enhanced Input and Comprehension	6
Reduced Speech-related Anxieties	6
Enhanced Vocabulary Acquisition Skills	8
Enhanced English Translation Skills	3
Enhanced English Listening Skills	1

This illustrates an increasing recognition within the literature for the potential use of AI within the context of EFL studies. The concentration of studies within the context of higher education illustrates a lag within the literature for the use of AI within the context of the K-12 or early childhood sectors. The systematic review illustrates a significant concentration within the literature for EFL learners within the context of higher education (75% of all studies), while elementary, secondary, and preschool students are strikingly absent from the literature. This mirrors the findings of prior systematic reviews (Sharadgah & Sa'di, 2022) but contributes to the ongoing discourse by connecting this phenomenon to a more significant theoretical disconnection. The identification of 103 relevant studies within a condensed six-year period illustrates the burgeoning nature of AI within the context of EFL studies; however, this also illustrates the nascent nature of this body of literature, including the demographic and methodological concentrations discussed within this review.

The dominance of research on adult learners is no coincidence, but rather an intrinsic relationship between current AI technology and the underpinning philosophy of andragogy (Knowles, 1984). Technologies such as automated writing evaluators are predicated on self-directed learners who are able to interact with abstract, text-based tasks and utilize feedback in a meta-cognitive fashion. Furthermore, from a sociocultural theory standpoint (Vygotsky, as cited in Hausfather, 1996), these systems are able to operate effectively as technology scaffolding in the Zones of Proximal Development of adult learners, in terms of supporting language production when designed to address complex, self-initiated queries.

However, such alignment also gives rise to a fundamental misalignment in terms of pedagogical paradigms that are considered necessary for young learners. The constructivist theory, as advocated by Piaget and Duckworth (1970), states that knowledge, particularly language, is constructed through concrete, sensory, and play-based interactions with the environment. The most widely accepted research paradigm in our field, which focuses on text-based and dialogue-based tools for the needs of autonomous learners, does not address such a vital form of engagement. While the sociocultural theory does address learning as a socially mediated process, such AI tools for young learners tend to act more as a form of transactional answer providers, such as image-recognition translators, instead of being able to act as collaborative learning partners that can provide the nuanced and progressive form of interaction necessary for a child to learn within their Zone of Development. Therefore, such a lack of research into young cohorts indicates a theory-to-practice gap that is not simply indicative of a lack of such applications but, indeed, a lack of AI systems that were designed from the outset based on our current understandings of child

development psychology and the socio-experiential nature of young child language acquisition. The limited studies focusing on younger learners, such as those using AI chatbots or translation apps to boost engagement and vocabulary acquisition (Hsu et al., 2023; Jeon, 2023; Lew et al., 2024), typically employ simpler, more guided tools—a practical choice that inadvertently highlights the gap in AI systems designed for developmental appropriateness.

In terms of research methodology, quantitative approaches dominate (62%), followed by mixed methods (32%) and qualitative designs (6%). Quantitative designs effectively quantify concrete outcomes, such as test score improvements, but may not adequately inform us about learning processes. Such a limitation is a theoretical limitation. From an interpretivist approach, learning is a process of meaning-making. Although dominant quantitative paradigms are effective in establishing objective correlations, they may not adequately inform us about how learners make sense of AI feedback or how such technologies impact motivational constructs such as autonomy, competence, and relatedness, which are core constructs of Self-Determination Theory (Ryan & Deci, 2000). Qualitative and mixed-methods research are therefore essential to interrogate these lived experiences and provide causal explanations for the outcomes quantified in dominant studies. Thus, while the steady increase in publications from 2019 to 2024 signals growing interest, the results of such an undertaking reveal that the way forward for the field is dependent upon expanding its methodological scope. Specifically, there is a need for more qualitative and longitudinal approaches to complement the dominant quantitative paradigm. It is only through such a diversified approach that it is possible to transcend issues of short-term efficacy and develop a theoretically and pedagogically sophisticated understanding of the role of AI in language education for all demographics of learners.

5.2. The most common AI technologies and AI-based tools used for EFL learning (RQ2)

The categorical analysis of AI technologies and tools carried out in this review (see Section 4.2 and Tables 2 and 3) shows that specific concentrations are evident in contemporary EFL studies. In particular, deep learning dominates with 51% of all core technologies identified in this review, while AI chatbots make up 56% of all tools analyzed. The dominance of these specific technologies cannot be explained by chance; instead, it points to a substantial match with what AI technologies can currently offer and with what theoretical paradigms underpin second language acquisition.

Firstly, the pervasiveness of AI chatbots, especially those based upon generative AI technology or more advanced models of NLP, underscores a significant research focus upon output-oriented models of interactive learning environments. The high volume of research studies (totaling 44 articles) suggests that such technologies have the ability to significantly facilitate learners' oral fluency development, as well as their willingness to communicate and writing output efficiency (e.g., Chen, Hsu, et al., 2022; Kim et al., 2021; Yuan, 2023). From a theoretical perspective, such an empirical finding supports the predictions of Sociocultural Theory and the Output Hypothesis. These chatbots can be seen as a form of "simulated interlocutors" or "scaffolding" that provide learners with a critical opportunity to produce output and negotiate meaning, which are essential in the development of automaticity in language use (Swain & Lapkin, 1995).

Second, the dominance of deep learning as a core enabling technology provides an explanatory foundation for the level of

sophistication found in these applications, which, in turn, supports the longstanding pedagogical concept of adaptive learning. This present review has shown that deep learning-based technologies, which include NLP, ASR, image recognition, etc., represent a broad range of learning tools, which vary from intelligent writing assistants like Grammarly (Dizon & Gayed, 2021; He, 2024; Liu, Hou, et al., 2023; Sanosi, 2022; Taskiran & Goksel, 2022; Wang et al., 2022), oral proficiency assessment tools like those described by Jiang et al. (2023) and Wang, D. (2024), to personalized vocabulary learning assistants like the one described by Yu (2023), among others. These learning tools have the capacity to analyze complex linguistic patterns, which substantively supports the concept of Adaptive Learning Theory, which, in turn, argues that learning design must be able to adapt to individual learning paths, needs, and achievement levels (Rachmad, 2022). For instance, the “English-intelligent tutoring teaching models” used in some of the studies reviewed, which are based on deep learning, have shown a level of success in improving the comprehensive achievement levels of the learners (Gao, 2021; Liu, 2023; Luo, 2024; Zhao, 2022).

By providing a detailed two-tier taxonomy of core AI technologies and applied tools, and by clearly delimiting the concept of AI chatbots as a specific form of applied generative AI, this review contributes to and clarifies existing systematic reviews of the field. For example, although a generally positive impact of AI on Learning Management Systems is noted by Oluwafemi Ayotunde et al. (2023), this review identifies the specific focus of recent research as being concerned with deep learning-based tools, particularly advanced chatbots and personalized assessment tools, as the driving factor behind such findings, indicating a notable development in the field which may not have been immediately apparent in earlier reviews. However, as a result of this congruence between theory and technology, a notable imbalance is revealed in the state of research and development in the field, as indicated by the findings summarized in Table 4, which show a marked predominance of studies on writing (20 studies) and speaking (18 studies) over listening and reading comprehension (only 7 studies combined). From the point of view of a comprehensive second language acquisition theory, such a predominance is notable, as the tools and technologies developed for productive skills are clearly aligned with the Output Hypothesis, while Krashen's (1989) Input Hypothesis emphasizes the importance of comprehensible input as a driving factor behind language acquisition, and yet the current state of the field suggests a notable bias towards the development of AI tools and technologies which are able to provide sophisticated support for language production, drawing on existing NLP and ASR techniques, while the development of AI tools and technologies which provide sophisticated support for language comprehension, such as deep listening comprehension and interpretive reading, is a relatively under-explored field, as indicated by the small number of studies on intelligent tutoring systems for reading (Xu et al., 2022, 2024) and subtitle technology for comprehension (Malakul & Park, 2023).

The educational implications of the aforementioned findings can be twofold. Firstly, the effectiveness of tools like AI chatbots and automated writing evaluation (AWE) tools can provide educators with a resource that can aid in the teaching of fluency and accuracy, as both are crucial in the teaching of fluency and accuracy. Secondly, the aforementioned findings can also have a negative educational implication, as an over-dependence on the most commonly used AI tools can lead to an instructionally biased approach that may prioritize the teaching of productive skills while neglecting the need to ensure that receptive skills are also being sufficiently addressed, possibly by using non-AI interventions or future AI tools that can aid in the teaching of listening and reading skills.

The current review, therefore, attempts to lay down specific guidelines for future research and development. Such developments should not be restricted to the development of more popular tools individually but should be geared towards designing intelligent learning environments that synergistically use deep learning techniques across all aspects of language learning, from intelligent input modification and planning

of learning paths to interactive output and multidimensional feedback. Specifically, there is a need to develop more AI tools that assist with listening and reading skills with the same degree of sophistication that is being given to writing and speaking. Furthermore, there is a need to conduct more comparative longitudinal studies to examine the effectiveness of such new forms of integrated learning systems compared to more traditional forms of learning and output-oriented learning tools.

5.3. The effects developed in research on AI technologies for EFL learning (RQ3)

With these dominant technologies established in RQ2, the learning outcomes analysis of RQ3 shows a largely positive research environment. However, a theoretical synthesis of these established findings shows a rich pedagogical potential and clearly defines the necessary direction of future research. The predominance of studies showing improvements in integrated skill, writing, and speaking abilities (see Table 4) clearly corroborates and expands on existing research findings. For example, while Oluwafemi Ayotunde et al. (2023) found a positive effect of AI on language learning within a Learning Management System environment, this study aggregates findings across a wider spectrum of technologies to clearly demonstrate that AI-based tools, specifically AWE systems and highly developed chatbots, support productive skill development. This convergence of findings clearly establishes a more solid empirical basis for the implementation of key second language acquisition theories. The effectiveness of such tools in enhancing writing and speaking abilities can be clearly understood within the framework of the Output Hypothesis (Swain & Lapkin, 1995) and the Noticing Hypothesis (Schmidt, 1990), where such tools provide a structured environment to facilitate “pushed output” and direct learners' attention to specific language forms to facilitate interlanguage development. Similarly, reductions in learning and speaking anxiety are clearly consistent with Krashen's (1989) Affective Filter Hypothesis, where such a low-anxiety environment may be a more conducive environment for language learning.

However, this consistency, coupled with the notable lack of negative repercussions, suggests a need for a more nuanced and critical approach, as informed by pedagogical and motivational theories. The marked disparity between the 38 studies on writing and speaking, as opposed to the 7 studies on listening and reading, suggests a notable bias towards the theoretical and technical aspects of the field. Such a bias suggests that, at present, AI applications are largely geared towards assessing and scaffolding language production, as informed by existing techniques and approaches to natural language and speech processing. The relative lack of attention to listening and reading comprehension suggests a notable gap, particularly as informed by comprehensive approaches to language proficiency, as underscored by the importance of comprehensible input (Krashen, 1989). As such, this marked bias is not merely a function of research priorities but a notable pedagogical gap, as the field has yet to develop AI tools geared towards scaffolding listening comprehension and interpretation at a level of sophistication commensurate with existing approaches to providing written language and speech production feedback. Furthermore, the overwhelming positive findings of studies on short-term efficacy prompt a consideration of these tools in relation to long-term learner development and autonomy within the framework of Self-Determination Theory (Ryan & Deci, 2000). Whereas tools such as AWE systems and chatbots may have a positive impact on learners' motivation and self-efficacy in the short term, as suggested by studies on Grammarly (Dizon & Gayed, 2021) and ChatGPT (Yuan, 2023), little is known about their potential impact on learners' intrinsic need satisfaction in autonomy, competence, and relatedness over time. Important questions remain unanswered with regard to whether excessive dependence on such tools may actually hinder the development of self-regulatory and self-editing learning strategies. Furthermore, with regard to pragmatic competence and sociolinguistic competence, it must be acknowledged that the potential of learners to internalize

pragmatically unnatural and contextually inappropriate language forms from interactions with such tools constitutes a serious concern that existing studies are not designed to investigate. This theoretical concern underscores a major methodological limitation in the extant literature in the predominance of short-term and quantitative studies, and highlights the pressing need for longitudinal, mixed-methods research.

Thus, while RQ3 offers robust evidence of the efficacy of AI in improving key productive skills and affective states, the above theory-driven discussion reveals key blind spots in the research and delineates key areas that must be taken into consideration in the development of AI in the future. Pedagogically speaking, a nuanced and critical approach must be taken in the use of AI. While AI systems must be used in a manner that is strategically aligned with a holistic and comprehensive curriculum that supports receptive skills as equally as productive skills, possibly through the development of AI systems that offer interactive listening and reading supports in the future. Educators must also strive to create a framework that encourages learners to use AI as a stimulus for meta-cognitive reflection and discussion with a human partner rather than encouraging dependency. For researchers, priorities must include developing AI technologies that offer sophisticated, adaptive support for listening and reading comprehension, designing and implementing longitudinal studies to examine the sustained impact of AI tool use on motivation, autonomy, and pragmatic competence, and employing richer, qualitative methodologies to understand the lived experience of learners within AI-mediated environments. By moving beyond a mere cataloguing of positive effects to engage deeply with pedagogical theory and learner development, the field can ensure that AI integration genuinely enriches and balances the complex process of language acquisition.

6. Conclusions and recommendations

This systematic review has mapped the landscape of research on AI technologies and AI-based tools in EFL learning from 2019 to 2024, addressing publication trends, participant demographics, prevalent technologies, and reported learning outcomes. Beyond cataloguing these trends, the discussion has interpreted the findings through key pedagogical and second language acquisition theories, revealing a field marked by both significant promise and critical imbalances. The primary conclusions are as follows. First, there has been a steady increase in publications, yet the research demographic is heavily skewed toward tertiary-level learners (75% of the reviewed studies). This concentration reflects a theoretical misalignment where prevalent AI tools, designed around principles of andragogy and textual autonomy, largely overlook the constructivist, socio-affective, and experiential learning needs of younger (K–12 and preschool) learners. Second, in terms of technological deployment, deep learning emerges as the foundational AI technology (51%), predominantly powering the most frequently researched tool category: AI-powered chatbots (i.e., those utilizing generative AI or advanced NLP models) (56% of tool-focused studies). These tools are theoretically aligned with the Output Hypothesis and Sociocultural Theory, providing valuable opportunities for interactive practice and feedback. However, this focus reveals a pronounced imbalance, with research and development heavily favoring writing and speaking (productive skills) over listening and reading comprehension (receptive skills), a disparity that challenges the principles of comprehensive language proficiency models emphasizing comprehensible input. Third, the consolidated evidence points to overwhelmingly positive outcomes, particularly for enhancing integrated skills, writing, and speaking, as well as reducing learning anxiety. While these findings corroborate earlier reviews regarding efficacy, our analysis, informed by Self-Determination Theory and the Noticing Hypothesis, raises critical questions about the long-term impact on learner autonomy, self-regulation, and the development of pragmatic competence that are not yet captured by the predominantly short-term, quantitative studies dominating the field. The contribution of this review is threefold. First,

unlike prior descriptive reviews, it offers a theory-driven interpretation of observed patterns by applying five explicit theoretical frameworks. Second, it introduces a methodological innovation by distinguishing between foundational AI technologies and applied AI-based tools through a two-tier taxonomy. Third, this theoretically grounded approach reveals critical imbalances in the field, such as the concentration on productive over receptive skills, the predominance of quantitative over qualitative methodologies, and the focus on higher education over K-12 learners, and generates specific, theory-informed recommendations for future research, pedagogical practice, and technological development.

These conclusions lead to several targeted recommendations for researchers, educators, and developers. For future research, a deliberate expansion of scope and methodological rigor is essential. Studies must move beyond the convenience sample of higher education to investigate theory-informed, age-appropriate AI applications for elementary, secondary, and preschool EFL learners. Furthermore, to move beyond measuring short-term efficacy, longitudinal and mixed-methods designs are urgently needed to examine the sustained impact of AI tools on motivational constructs (autonomy, competence, relatedness), learner self-regulation, and the development of higher-order skills such as critical thinking and pragmatic competence. Comparative studies that evaluate the efficacy of emerging integrated intelligent learning environments against both traditional methods and standalone tools would also be valuable. For pedagogy and instructional design, a balanced and critical integration strategy is paramount. Educators should leverage the proven benefits of tools like AI chatbots and automated writing evaluators while consciously ensuring that receptive skill development receives equal attention, potentially through complementary non-AI or future AI-driven scaffolds. Crucially, AI tools should not be transferred uncritically across age groups; applications for younger learners must be designed from a developmental psychology perspective. Pedagogical activities should be designed to use AI-generated feedback as a starting point for metacognitive reflection and human dialogue, thereby fostering learner autonomy rather than dependency. For technological development, the current imbalance presents a clear directive. There is a pressing need for innovation in AI tools that offer sophisticated, adaptive support for listening and reading comprehension, matching the level of investment currently dedicated to productive skills. Future tools should aim to create integrated intelligent learning environments that synergistically support the full spectrum of language acquisition, which are from intelligent input processing and personalized learning paths to immersive interaction and multidimensional assessment.

Acknowledging the limitations of this review, we recommend that future systematic reviews in this domain expand their search to include multiple databases (e.g., Web of Science, ERIC) to mitigate potential publication bias and capture a broader spectrum of literature. Finally, as the field matures, dedicated investigation into the potential side effects, limitations, and ethical considerations of AI in language learning, such as impacts on academic integrity, critical thinking, and the perpetuation of biases, is strongly recommended to provide a more holistic and objective understanding for all stakeholders.

CRedit authorship contribution statement

William Ko-Wai Tang: Writing – review & editing, Validation, Supervision, Resources, Investigation, Formal analysis, Conceptualization.
Xue-Ying Zhang: Writing – original draft, Conceptualization.

Ethics

This review does not involve any human participants. This review followed the PRISMA guidelines for transparency and comprehensive reporting and respected all copyrights and intellectual property rights associated with the studies.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. The data extracted from the 103 articles

ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
Enhanced integrated English skills						
1	Ying (2023)	Cross-levels	Mixed method	N/A	AI chatbot	Participants' language proficiency, motivation, capacity for critical thought, and overall learning experience are all enhanced by technology.
2	Sun and Li (2020)	Tertiary education	Quantitative method	Machine learning (Non Deep Learning)	N/A	Technology can assist students to improve their English language learning concepts, techniques, and topics while also igniting their interest and initiative.
3	Zhou (2023)	Tertiary education	Quantitative method	N/A	Computer-aided learning technology	With technology, students' post-assessment performance in English increased by 14.7%.
4	Liu (2023)	Tertiary education	Quantitative method	DL (NLP and recognition)	N/A	With technology, AI teaching effectively improves the students' business English scores.
5	Zhang (2023)	Tertiary education	Quantitative method	Model	N/A	With technology, Students' practical abilities, translation abilities, writing abilities, speaking abilities, and reading abilities have improved.
6	Kim (2022)	Tertiary education	Quantitative method	N/A	Computer-aided learning technology	Students' TOEIC scores were positively impacted by AI technology.
7	Zhao (2022)	Tertiary education	Quantitative method	DL(NN)	Intelligent tutoring system	With technology, the average score of students' monthly test tends to rise as a whole.
8	Gao (2021)	Tertiary education	Quantitative method	DL(RNN)	N/A	Technology can effectively promote the English scores of vocational college students.
9	Qianjing and Lin (2021)	Tertiary education	Quantitative method	Spectrum algorithm	Computer-aided learning technology	Technology can virtually improve students' English scores.
10	Nikonova et al. (2023)	Tertiary education	Quantitative method	N/A	Computer-aided teaching technology	Technology improves students' learning activities and motivation, which benefits their overall performance and lexical skills.
11	Wu and Wang (2021)	Tertiary education	Quantitative method	N/A	Computer-aided teaching technology	With technology, students' average English self-management learning skill grew by 11.42 points using technology, and their overall professional English score climbed by 8.16 points on average.
12	Xin (2021)	Secondary education	Quantitative method	N/A	Computer-aided teaching technology	Technology not only promotes students' academic performance but also enhance the students' comprehensive language ability.
13	Dillon (2024)	Tertiary education	Mixed method	N/A	AI chatbot	ChatGPT supports various learning strategies, which enhance linguistic competence and promote self-regulated learning.
14	Tu (2024)	Tertiary education	Quantitative method	N/A	Speech/image recognition system	The performance of the multimedia group improved significantly after the experiment, showing that intelligent multimedia image technology-based online English teaching is conducive to improving students' academic performance.
15	Dhanapal et al. (2024)	Unspecified	Mixed method	N/A	AI chatbot	The study shed light on the impact of AI-based training on language-learning outcomes.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
16	Liu and Wang (2024)	Tertiary education	Mixed method	Deep Learning (NLP)	N/A	Students in the experimental class performed significantly better than those in the control class on various English skill tests, with notable improvements in listening, reading comprehension, and business English writing.
17	Yan J. (2024)	Tertiary education	Quantitative method	Model	N/A	The learning ability significantly increased from a score of approximately 57 points to 78 points after the experiment.
18	He (2024)	Tertiary education	Quantitative method	Model	N/A	The experimental class outperformed the control class in course learning level, cooperative communication, pre-study, and classroom learning status.
19	Ouyang et al. (2024)	Tertiary education	Quantitative method	N/A	Translate app	Compared to the control group, the experimental group showed statistically significant improvements in willingness to speak, read, write, comprehend, and communicate.
20	Yan X. (2024)	Secondary education	Mixed method	N/A	Computer-aided learning technology	The AI tutoring model significantly improved students' language learning outcomes, with an average performance increase of 12.02 points, 10.19 points higher than the control group.
21	Ma and Chen (2024)	Tertiary education	Quantitative method	Deep Learning(NN)	N/A	The experimental group using AI technology saw an overall score increase of 12.33 points, with a P-value of less than 0.01 across four dimensions. Their average comprehensive score in English assessment was 95 points, 8 points higher than the control group using traditional methods.

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ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
22	Yu and Wang (2024)	Tertiary education	Quantitative method	Model	N/A	Students using the AI system with an average score above 60 maintained or improved their grades. Their engagement level was positively correlated with grade improvement.
23	Wang, D. (2024)	Tertiary education	Quantitative method	Model	N/A	The blended teaching mode outperformed traditional methods by 0.33 points in entrance exams, 2.96 points in Grade 4 total scores, 2.10 points in listening, 0.57 points in reading, and 0.31 points in writing. The effect was positively correlated with students' initial English proficiency.
24	Liu et al. (2024)	Tertiary education	Mixed method	Model	N/A	The experimental group scored 4.2 points higher than the control group, with a 100% passing rate and most scores clustering between 70 and 80 points. AI teaching significantly improved business English scores.
25	Luo (2024)	Tertiary education	Quantitative method	Deep Learning (RNN, CNN, and NLP)	N/A	After 12 h hours of AI-assisted precision education, students' proficiency in listening, speaking, reading, writing, and translating improved significantly, enhancing overall college English proficiency.
Enhanced English writing skills						
26	Wang (2021)	Tertiary education	Mixed method	N/A	N/A	With technology, the students' independent learning ability and English writing ability were significantly improved.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
27	Liu, Hou, et al. (2023)	Tertiary education	Mixed method	N/A	Automatic evaluation/feedback system	Technology not only greatly enhanced the English writing performance of students, but it also significantly decreased their cognitive load, increased their self-efficacy, and promoted self-regulated learning.
28	Wang et al. (2022)	Tertiary education	Quantitative method	N/A	Automatic evaluation/feedback system	Technology remarkably enhanced the students' writing performance, critical thinking, self-efficacy, sense of presence, as well as learning motivation. It also decreased their anxiety related to EFL writing.
29	Liu, Liu, et al. (2023)	Tertiary education	Mixed method	N/A	Automatic evaluation/feedback system	Students' use of technology enhanced their writing abilities, increased their drive to learn, sharpened their critical thinking skills, and greatly decreased their writing anxiety.
30	Sanosi (2022)	Tertiary education	Quantitative method	N/A	Automatic evaluation/feedback system	With technology, the experimental group members outperformed the control group in terms of writing accuracy. Additionally, the researcher discovered a significant decrease in the quantity of mistakes in English composition.
31	Taskiran and Goksel (2022)	Unspecified	Quantitative method	N/A	Automatic evaluation/feedback system	Technology can be considered as an affordable means of editing and improving students' written work. It could also encourage students to work hard to complete the assigned tasks and maintain their motivation over time.
32	Yan (2023)	Tertiary education	Qualitative method	N/A	AI chatbot	Technology could maximize the efficiency in composing writing.
33	Dizon and Gayed (2021)	Tertiary education	Quantitative method	N/A	Automatic evaluation/feedback system	With technology, students made fewer grammatical mistakes and used a wider variety of words when writing.
34	Xie (2022)	Unspecified	Quantitative method	N/A	Computer-aided teaching technology	With technology, the number of words memorized by students increases and they are making less mistakes while writing in English.
35	Li et al. (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	The study highlighted that ChatGPT significantly enhanced the English academic writing proficiency of non-native medical students and showed potential in contributing to the educational evaluation process in non-English-speaking environments.
36	Teng (2024)	Tertiary education	Mixed method	Deep Learning (NLP)	AI chatbot	The findings supported the significant positive effects of AI assistance on writing, including improvements in writing motivation, self-efficacy, engagement, and collaborative writing tendency among EFL learners.
37	Tsai et al. (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	Significant improvements were observed in all four dimensions of writing quality assessment (vocabulary, grammar, organization, and content) for EFL college English majors who used ChatGPT-assisted revision, although concerns about fairness in evaluation were also raised.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
38	Guo et al. (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	The intervention significantly improved students' feedback quality and enhanced the writing skills of feedback providers.
39	Bacon and Maneerutt (2024)	Tertiary education	Mixed method	Deep Learning (NLP)	AI chatbot	The results showed substantial improvements in writing scores, as well as increased student confidence and engagement.
40	He (2024)	Tertiary education	Quantitative method	N/A	Automatic evaluation/feedback system	Participants in the experimental group performed better than those in the control group with respect to motivation to write, enjoyment in writing, academic buoyancy, and academic success in writing.
41	Muslimin et al. (2024)	Tertiary education	Mixed method	N/A	AI chatbot	The implementation of Cami AI-SAMR had a significant impact on EFL students' writing achievement.

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ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
42	Rahmi et al. (2024)	Tertiary education	Qualitative method	N/A	AI chatbot	The AI writing assistant helped correct grammatical errors and enhance text cohesion and coherence, but sometimes lacked content density and included unrelated ideas.
43	Polakova and Ivenz (2024)	Tertiary education	Mixed method	N/A	AI chatbot	Significant improvements in writing aspects, such as conciseness, grammar, inclusion of key information, and use of passive voice, were observed following ChatGPT feedback.
44	Wang, D. (2024)	Tertiary education	Mixed method	N/A	AI chatbot	Quantitative analysis showed that ChatGPT feedback significantly enhanced various aspects of writing, such as conciseness, grammar, inclusion of key information, and use of passive voice.
45	Chen et al. (2024)	Secondary education	Mixed method	Deep Learning (RNN, CNN, and NLP)	N/A	Independent-sample and paired-sample t-tests confirmed that the proposed system significantly enhanced students' EFL writing proficiency.
Enhanced English speaking skills						
46	Young and Wang (2014)	Others	Quantitative method	N/A	AI chatbot	Technology's dialogues are well suited for students and easily comprehensible, enabling students at the elementary level and intermediate level to hold most of the vocabulary used.
47	Jiang et al. (2023)	Tertiary education	Quantitative method	DL (ASR)	Speech/image recognition system	With technology, significant gains in phonological accuracy, speed fluency, repair fluency, and lexical and morphosyntactic accuracy were witnessed.
48	Yuan (2023)	Elementary education	Quantitative method	N/A	AI chatbot	Technology greatly enhanced students' oral English proficiency and willingness to communicate.
49	Chen, Koong, and Liao (2022)	Elementary education	Mixed method	N/A	Speech/image recognition system	Technology can effectively help students improve their English-speaking abilities and lessen their fear of learning the language.
50	Kim et al. (2021)	Tertiary education	Quantitative method	N/A	AI chatbot	With technology, most of the students were able to speak more proficiently after communicating with AI. However, the low-level group did not show considerable improvement in fluency after practicing speaking with AI.
51	Rahman and Tomy (2023)	Tertiary education	Quantitative method	NLP	N/A	Technology was a useful and suitable tool for enhancing the oral fluency of L2 learners.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
52	Chen, Hsu, et al. (2022)	Tertiary education	Qualitative method	N/A	AI chatbot	With technology, the students were able to improve their fluency, speaking abilities, and vocabulary.
53	Sayed et al. (2024)	Tertiary education	Mixed method	Deep Learning (NLP)	AI chatbot	The results indicated notable increases in speaking abilities, psychological well-being, learner autonomy, and academic buoyancy among EFL learners who used AI-powered tools, suggesting that AI systems, like ChatGPT, can transform language assessment and pedagogy by improving skill development and meeting students' emotional and psychological needs.
54	Qassrawi et al. (2024)	Tertiary education	Mixed method	N/A	AI chatbot	The results indicated substantial improvements in writing scores and increased student confidence and engagement among EFL learners who used ChatGPT for writing feedback in combination with peer-assisted learning.
55	Yoo and Ahn (2024)	Elementary education	Quantitative method	Deep Learning (NLP and Recognition)	AI chatbot	Prosody training with pitch contour feedback is a promising method for improving English pronunciation among young Korean EFL learners.
56	Dennis (2024)	Tertiary education	Mixed method	N/A	AI chatbot	The technology was found to be effective in improving EFL learners' pronunciation and speaking skills.
57	Duong and Suppasetseree (2024)	Tertiary education	Mixed method	N/A	AI chatbot	Students' English speaking skills improved significantly after practicing with an AI voice chatbot.
58	Mohammadkarimi (2024)	Unspecified	Mixed method	N/A	AI chatbot	AI-based pronunciation tools can significantly improve pronunciation accuracy by providing individualized feedback and focused practice opportunities.
59	Du (2024)	Tertiary education	Quantitative method	Model	N/A	The GRU algorithm played a crucial role in extracting English language and literature features, with a significant reduction in student speech error rates from approximately 2% to 1.21%.
60	Wang (2024)	Unspecified	Quantitative method	Deep Learning (Recognition)	Speech/image recognition system	Real-time, adaptive feedback significantly improved pronunciation accuracy, reduced error rates, and refined articulation precision among Chinese learners of English.
61	Mirzoyeva et al. (2024)	Tertiary education	Mixed method	Deep Learning (NLP)	AI chatbot	Pre- and post-test evaluations revealed significant improvements in auditory and speaking skills for both groups, demonstrating the effectiveness of the course.
62	Yuan (2023)	Elementary education	Mixed method	N/A	AI chatbot	Chatbot integration significantly improved oral English proficiency and willingness to communicate in the experimental group compared to the control group.
63	Kemelbekova et al. (2024)	Tertiary education	Mixed method	N/A	AI chatbot	Both groups showed significant improvement in oral communication skills, as shown by their performance in reading aloud and answering questions.
Enhanced cognition						
64	Ni and Cheung (2023)	Secondary education	Quantitative method	N/A	Intelligent tutoring system	Perceived usefulness, financial value, learning goal orientation, supportive conditions, and students' actual usage

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ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
of technology for English learning were all positively correlated with their intention to continue using it.						
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
65	Zhang et al. (2023)	Tertiary education	Quantitative method	N/A	AI chatbot	Students' understanding of technologies has a significant correlation with their language learning achievements and it has a major moderating effect on the relationship between knowledge and outcomes.
66	Liu and Li (2022)	Secondary education	Quantitative method	N/A	Computer-aided learning technology	Classes that have adopted the technology demonstrate higher language cognitive abilities, which is especially more useful for beginners.
67	Mohamed (2024)	Others	Qualitative method	N/A	AI chatbot	While some academics recognized the technology's value in offering prompt and accurate answers to a variety of queries, others voiced worries that it may impede students' ability to think critically and do independent research as well as perpetuate prejudices and false information.
68	Liu and Wang (2024)	Tertiary education	Mixed method	N/A	AI chatbot	The experimental group showed a statistically significant improvement in critical thinking skills compared to the control group, indicating that AI tools can effectively enhance critical thinking abilities in English literature classes.
Enhanced input and comprehension						
69	Srinivasan and Murthy (2021)	Cross-levels	Quantitative method	N/A	Computer-aided learning technology	With technology, children's English reading and comprehension skills are improved with an increase of 20–40%.
70	Leng and Lin (2024)	Tertiary education	Quantitative method	Model	N/A	With technology in teaching, students' comprehensive strengths and the cultivation of high-quality English talent for universities have been improved.
71	Malakul and Park (2023)	Secondary education	Mixed method	DL (ASR)	N/A	Technology can facilitate students' learning comprehension, cognitive load, and satisfaction.
72	Xu et al. (2022)	Tertiary education	Quantitative method	N/A	Intelligent tutoring system	Technology improved Chinese EFL students' reading comprehension significantly.
73	Chung and Bong (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	L2 pronunciation training using AI apps significantly improved L2 perceptual competence and increased L2 learning motivation.
74	Xu et al. (2024)	Tertiary education	Quantitative method	N/A	Intelligent tutoring system	The ITSS intervention had a statistically significant positive effect on adult Chinese English Language Learners' reading comprehension.
Enhanced English vocabulary acquisition skills						
75	Jeon (2023)	Elementary education	Quantitative method	N/A	AI chatbot	Technology could enhance vocabulary acquisition and provide diagnostic information about learners.
76	Hsu et al. (2023)	Elementary education	Quantitative method	DL (AIR)	Speech/image recognition system	Technology was helpful in terms of vocabulary acquisition, self-regulated learning methodologies, and anxiety-reduction.
77	Yu (2023)	Tertiary education	Quantitative method	Machine learning (Non Deep Learning)	N/A	With technology, effective resources that are recommended for user groups with varying requirements, interests, and skill levels increase the effectiveness of learning vocabulary in English.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
78	Lew et al. (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	ChatGPT was found to be more effective than either dictionary in production tasks, and better than the monolingual dictionary in reception tasks, although not superior to the bilingual dictionary in reception.
79	Algraini (2024)	Tertiary education	Mixed method	N/A	AI chatbot	Participants reported that ChatGPT could provide accurate definitions, explanations, and examples. Its ability to offer contextual alternatives was also seen as useful for EFL students to choose appropriate words for specific contexts.
80	Ptasznik et al. (2024)	Tertiary education	Quantitative method	Deep Learning (NLP)	AI chatbot	ChatGPT outperformed in both vocabulary task reception and production. It was also more efficient in terms of consultation speed in the production task.
81	Wang et al. (2024)	Tertiary education	Quantitative method	Model	Computer-aided learning technology	AI-powered language learning platforms positively impacted English vocabulary learning outcomes, and the combined use of AI-powered mobile applications for self-testing and in-class tests effectively enhanced vocabulary learning efficiency.
82	Wang and Huang (2024)	Tertiary education	Quantitative method	Model	N/A	Personalized vocabulary recommendations were more effective than random selection in improving college students' English vocabulary learning.
Reduced speech-related anxieties						
83	Peng and Macwes (2023)	Unspecified	Quantitative method	DL (RNN, CNN, and NLP)	N/A	With technology, the assessment of foreign language anxiety is highly accurate, with an accuracy of 99% for the classification.
84	El Shazly (2021)	Tertiary education	Quantitative method	N/A	AI chatbot	With technology, FL learners had FLA both before and after the intervention. The determined anxiety levels contributed to FL learning in positive ways. Nevertheless, learners' anxiety connected to speaking did not decrease.
85	Chen, Koong, and Liao (2022)	Tertiary education	Qualitative method	N/A	Others	The instructional feedback of the technology decreased the participants' public speaking anxiety.

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ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
86	Tai and Chen (2024)	Elementary education	Mixed method	N/A	AI chatbot	CoolE Bot helped build rapport and a supportive environment, boosting learners' confidence and reducing anxiety related to EFL speaking.
87	Yildiz (2024)	Tertiary education	Mixed method	N/A	AI chatbot	Interviews indicated that engaging with ChatGPT helped learners build confidence in conversational skills, manage stress, and enjoy the learning process. AI chatbots like ChatGPT created a supportive environment that boosted speaking confidence.
88	Chen (2024)	Tertiary education	Mixed method	N/A	AI chatbot	PSA levels decreased in all groups after instruction, with the VR-facilitated group showing statistical significance. However, no differences were found in final oral performance across groups.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
Being willing to engage in EFL learning						
89	Zhang and Qi (2023)	Tertiary education	Quantitative method	Machine learning (Non Deep Learning)	Computer-aided learning technology	Technology can offer a comprehensive English language recognition system that satisfies the requirements of educators and learners, and its application in the classroom can help to establish a more effective, efficient, and customized learning environment for both parties.
90	Liu and Chen (2023)	Elementary education	Quantitative method	N/A	Translate apps	Young children could benefit from using the technology as a teaching tool since it increases students' interest in and mastery of foreign languages.
91	Ayedoun et al. (2020)	Tertiary education	Quantitative method	N/A	Intelligent tutoring system	With technology, learners tended to have higher expectations for their readiness to communicate.
92	Ibna Seraj and Oteir (2022)	Tertiary education	Mixed method	N/A	AI chatbot	Students' critical thinking can be enhanced as they become more trustful and self-assured, impartial, and sensible.
93	Ma and Chen (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	The experimental group exposed to AI tools showed significantly higher levels of affective, cognitive, and behavioural engagement compared to the control group, and also experienced a substantial reduction in academic procrastination.
94	Wiyaka et al. (2024)	Secondary education	Quantitative method	N/A	AI chatbot	Engagement with AI chatbots significantly alleviated reading anxiety and improved reading performance among Indonesian secondary school students, fostering a conducive learning atmosphere that enhanced confidence and engagement in acquiring the English language.
95	Ismail and Alharkan (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	The study found that using ChatGPT as an AI tool effectively improved EFL learners' enjoyment, self-efficacy, and resilience, with the experimental group outperforming the control group in all three post-tests.
96	Alsager (2024)	Tertiary education	Quantitative method	N/A	AI chatbot	Using ChatGPT as an AI learning tool reduced demotivation and foreign language learning anxiety among Saudi Arabian EFL learners while increasing their autonomy.
97	Yamaoka (2024)	Tertiary education	Qualitative method	N/A	AI chatbot	Students were motivated and experienced decreased anxiety. However, ChatGPT could reduce students' confidence if not used appropriately.
98	Jiang (2024)	Unspecified	Quantitative method	Model	N/A	AI technology accounted for 78% of the performance improvement in the English distance education system.
99	Pai et al. (2024)	Tertiary education	Mixed method	Model	N/A	Experimental class students showed significantly higher interest (47.44 ± 7.15) compared to the control class (29.65 ± 8.21). The system's multimodal display enhanced emotional performance, with virtual character emotions influencing identity and persuasiveness. Bidirectional interaction feedback made language learning more natural.
ID	Supporting Studies	Participants' educational level	Research Method	AI technologies	AI-based tools	Effect Description
Enhanced English translation skills						
100	Kim and Cha (2023)	Tertiary education	Quantitative method	N/A	Translate apps	Students' translations were perceived to be of significantly higher quality in terms of sentence structure. It is more advantageous to comprehend the entire chapter and to grasp its key notion. However, the research also had lower mean scores for interest and motivation for language acquisition.
101	Li and Shu (2022)	Tertiary education	Quantitative method	N/A	Automatic evaluation/feedback system	Students' translation skills can be improved by teacher input in addition to online automatic feedback.
102	Zhang (2023)	Tertiary education	Mixed method	Model	N/A	Students reported high satisfaction with personalized teaching (0.85) and evaluation (0.8). Translation ability improved from 75 to 90 under the English Intelligent Assisted Teaching Model.
Enhanced English listening skills						
103	Wang, J. (2024)	Tertiary education	Quantitative method	Model	N/A	After adopting the intelligent English listening training model, students in the observation group saw an 8.21-point increase in English listening scores compared to the control group.

Data availability

The data from this review is accessible to all interested parties upon reasonable request while respecting the privacy and confidentiality agreements of the original studies.

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